

ENV 6106

Air Pollution Modeling

Spring 2023

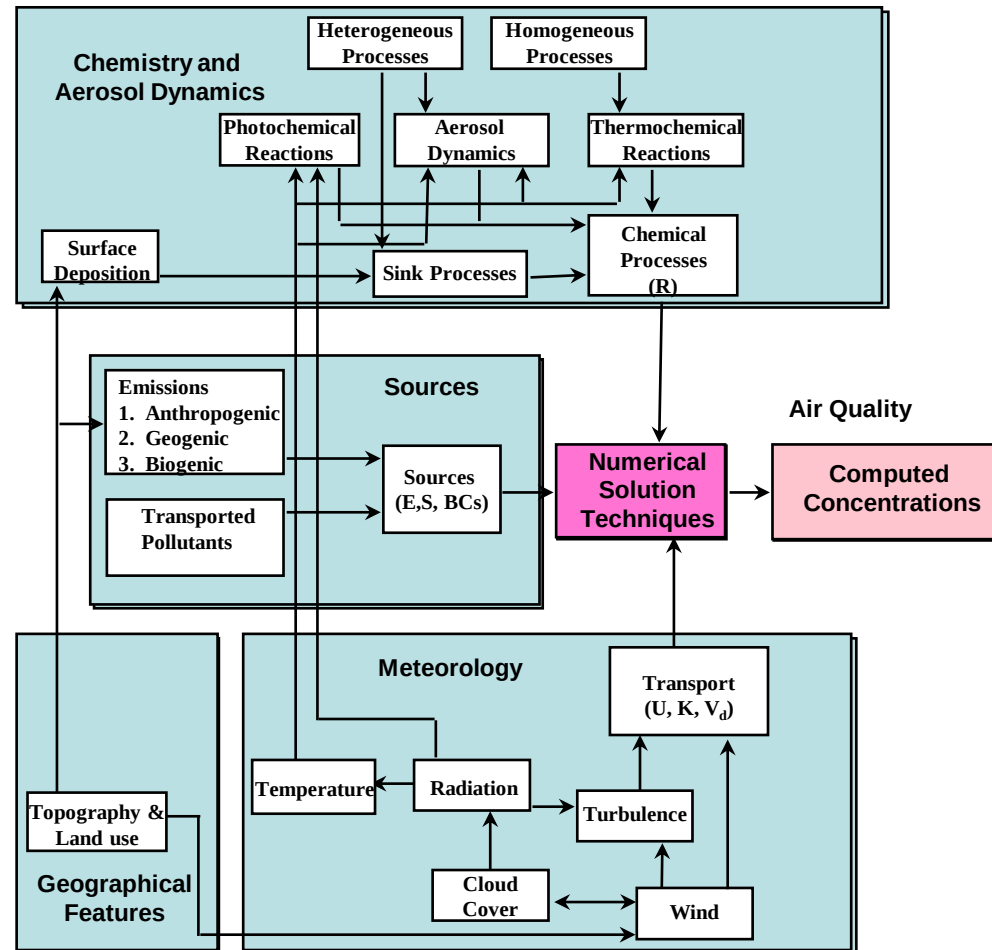
Topics

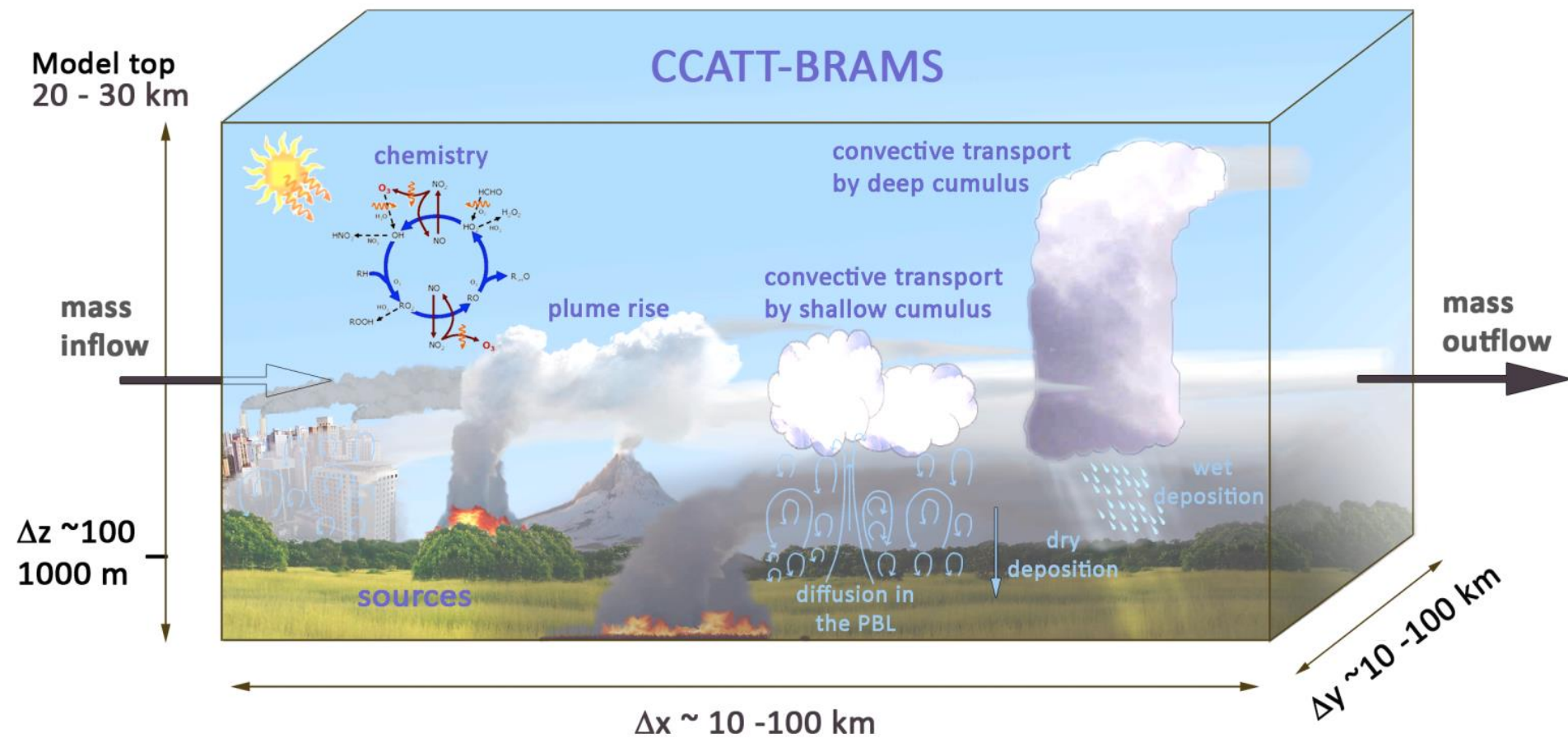
- Introduction to advanced air quality models
- Expectation
 - Understand, very generally, how chemistry, aerosol and emissions are handled in advanced air quality models

Four “layers”

- Functionally, an advanced modern air quality model is comprised of four layers
 - Physical layer: a simplified physical model to represent actual processes (with assumptions)
 - Math layer: mathematical representation of the physical layer (with more assumptions)
 - Numerical layer: numerical solution to math layer
 - Code layer: computer code for numerical layer
- Not an accurate representation

General framework of a modern air quality model





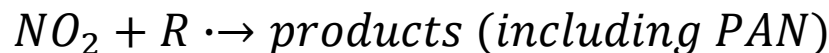
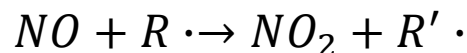
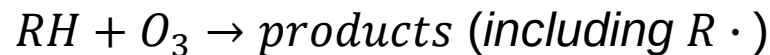
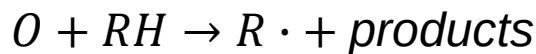
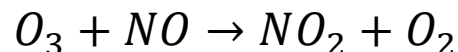
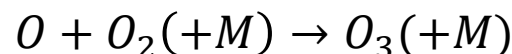
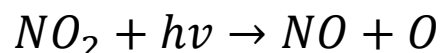
Trace gas sub-grid processes for in the CCATT-BRAMS model
 Lango et al., 2013 (doi:10.5194/gmd-6-1389-2013)

CMAQ: <https://www.epa.gov/cmaq/overview-science-processes-cmaq>

Chemistry

Chemistry in air quality models

- Numerous reactions occur in the ambient environment
- Different problem focuses on different groups of reactions, which requires detailed information on reactants/products



An early photochemical smog chemistry mechanism
(Friedlander and Seinfeld, 1969)

Doi: 10.1021/es60034a003

Chemistry in air quality models

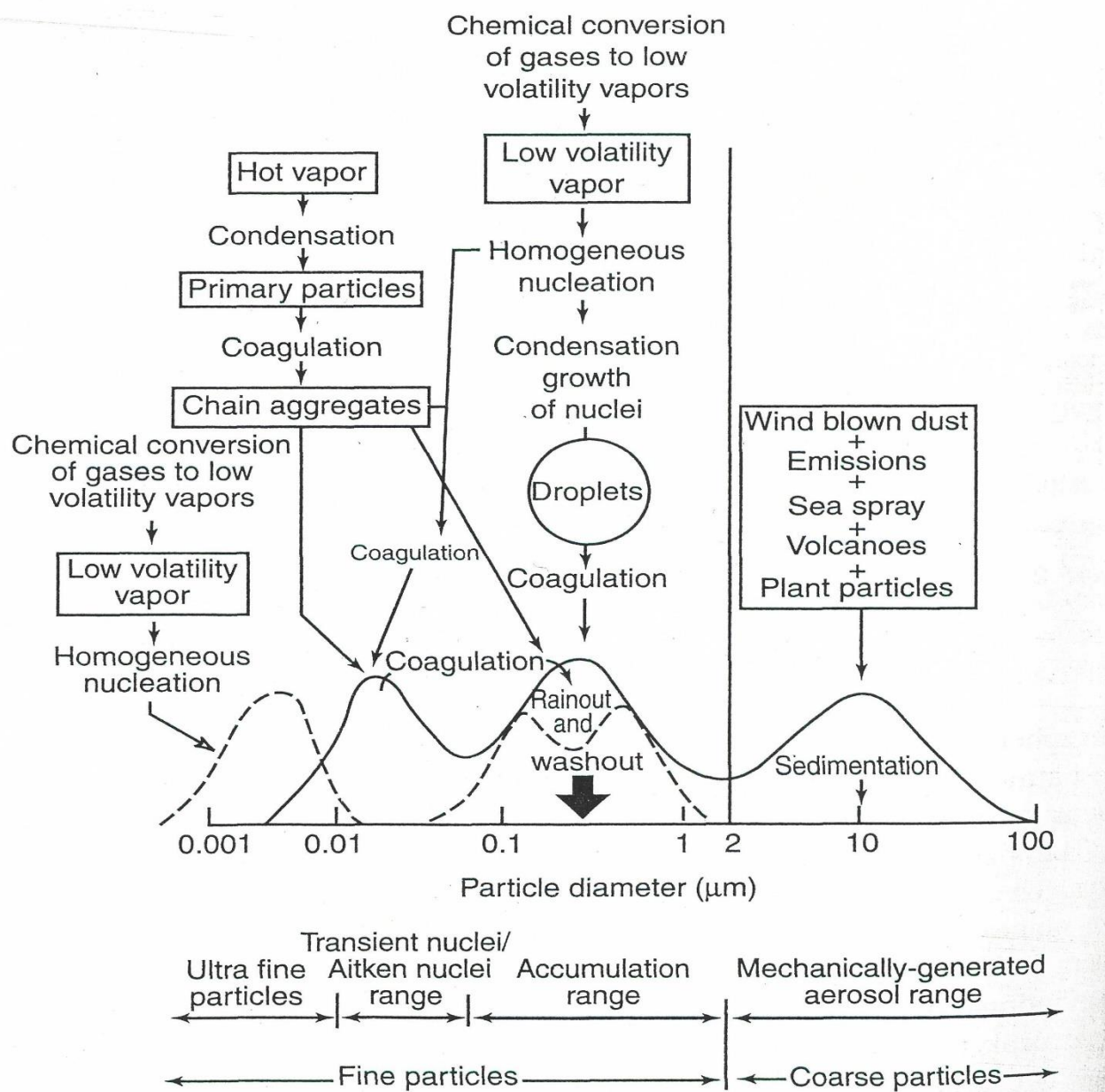
- You can choose to explicitly account for detailed reactions for specific chemicals
 - E.g: Master Chemical Mechanism (MCM); GATOR-GCMOM
 - Homepage: <http://mcm.york.ac.uk/home.htm>
 - V3.3.1 includes 17000 elementary reactions of 6700 primary, secondary and radical species
- Or you can choose to simplify things by lumping (aggregating) chemicals together
 - E.g: CB: Carbon Bond (CB); SAPRC (Statewide air pollution research center); RADM2 (2nd Regional acid deposition model mechanism); MPRM (Morphecul photochemical reaction mechanism)
 - SAPRC homepage: <https://intra.engr.ucr.edu/~carter/SAPRC/>

Chemistry in air quality models

- The complexity of chemical reaction mechanisms varies greatly
- Different mechanisms require information on different species
 - Concentration, environmental condition etc
 - Difficult to obtain all input data
 - “Speciation” in emission model: split one chemical into multiple ones

Aerosol process

Some aerosol processes



Source and processed of atmospheric aerosols at different size modes (Godish 2003).
 Solid line: three mode model by Whitby & co-worker; Dash line: added ultrafine mode
 with sometime observed accumulation model distribution. Vertical axis: maybe number,
 area or mass. This figure is for illustration only

Aerosol

- Dynamic model: physical & chemical processes that change particle concentration and size distribution
 - E.g: Nucleation, condensation, evaporation, coagulation, removal by precipitation etc
- Thermodynamic model: determine equilibrium concentrations of multiple species among gas, aqueous and solid phases
 - E.g: SO_4^{2-} — NO_3^- — NH_4^+ — Na^+ — Cl^- — H_2O
 - E.g: ISORROPIA
 - Home: http://nenes.eas.gatech.edu/ISORROPIA/index_old.html

Emissions

The NEI

- USEPA maintains the National Emissions Inventory (NEI), a “comprehensive and detailed estimate of air emissions of criteria pollutants, criteria precursors, and hazardous air pollutants”
 - Home: <https://www.epa.gov/air-emissions-inventories>
- Extensive collaborative efforts between USEPA and local agencies
- Updated every three years
- Do check this: <https://gispub.epa.gov/neireport/2017/>

The NEI

- NEI data collected/estimated using both top-down and bottom-up methods
 - Top-down: estimate total emission in an area
 - E.g: emissions from dry cleaning facilities in Orange County
 - $E = A \times EF$: Emission = Activity X Emission Factor
 - Available for the area of focus for an entire time (a county in a year)
 - Bottom-up: estimate/collect emission amount at individual source
 - E.g: Continuous Emission Monitoring System (CEMS) data at stacks
 - Can at specific location and very detailed

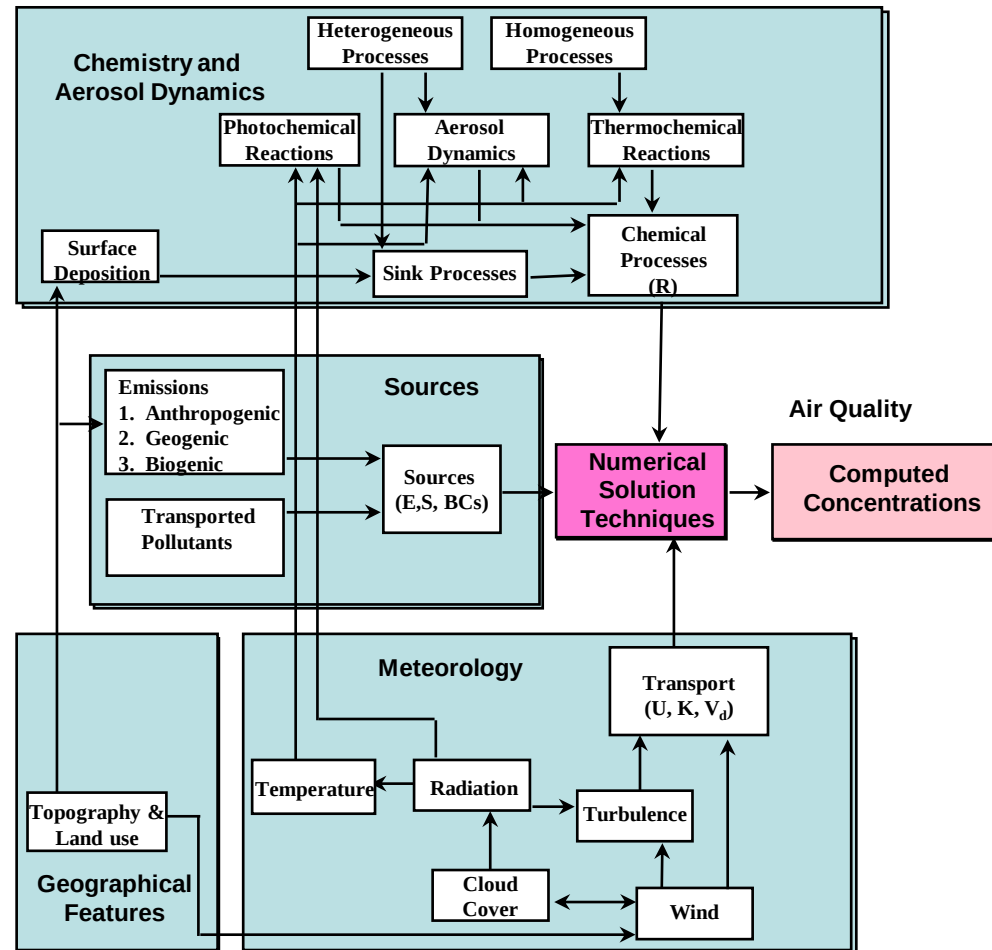
SMOKE

- SMOKE (Sparse Matrix Operator Kernel Emissions) is a widely used emission model in the US
 - Home: <https://www.cmascenter.org/smoke/>
- Prepare emission data for air quality models
 - Inputs in the US includes NEI
 - Also do speciation for different chemistry mechanisms
- Types of emission handled by SMOKE
 - Point, non-point (area), biogenic, mobile, others
- When only total emission is known
 - Spatial surrogate: used to spatially allocate emission
 - Temporal profile: used to temporally allocation emission

SMOKE emission processing

- Point: Many are detailed, may need temporal allocation
- Area: Spatial and temporal allocation
- Mobile: MOVES (Motor Vehicle Emission Simulator)
 - We will learn more later in this class
- Biogenic: BEIS (Biogenic Emission Inventory System) or MEGAN (Model of Emissions of Gases and Aerosols from Nature)

General framework of a modern air quality model



Reiterate

- *“all models are wrong”, but useful*
- There are always errors and biases in model outputs
- Caution must be taken to evaluate model outputs, and to identify and correct biases and errors
 - Data assimilations etc

Guideline

EPA guideline on air quality modeling

- EPA guideline is very detailed, and focuses mostly on point sources
- The guideline specific models for specific applications, and general modeling approaches
 - Permit
 - State Implementation Plan (SIP)
 - Screening
- URL: <https://www.epa.gov/scram>

History of the legal battle on GEP:

https://openscholarship.wustl.edu/cgi/viewcontent.cgi?referer=https://www.google.com/&httpsredir=1&article=1275&context=law_urbanlaw